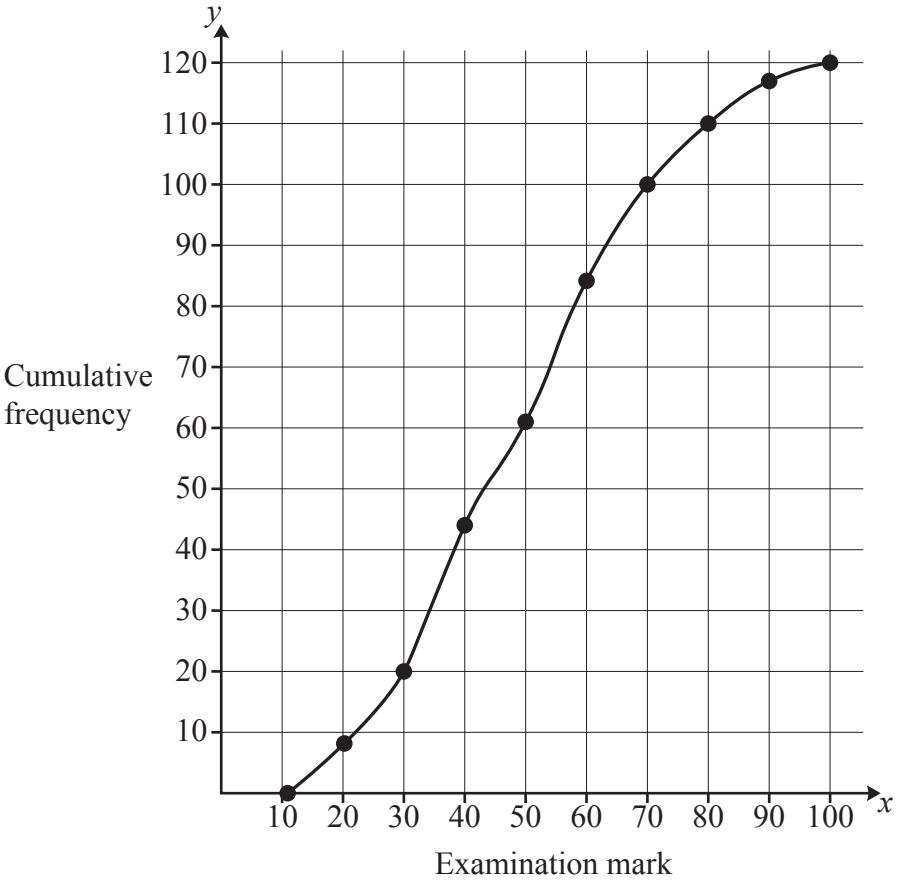
8 At the end of the summer term all 120 Year 8 pupils at Oakmount College sit the same mathematics examination.

The marks obtained by the pupils in the 2024 examination are summarised in the table.

Mark	11–20	21–30	31–40	41–50	51–60	61–70	71–80	81–90	91–100
Frequency	8	12	24	17	23	16	10	7	3

Software is used to produce the cumulative frequency curve shown below.



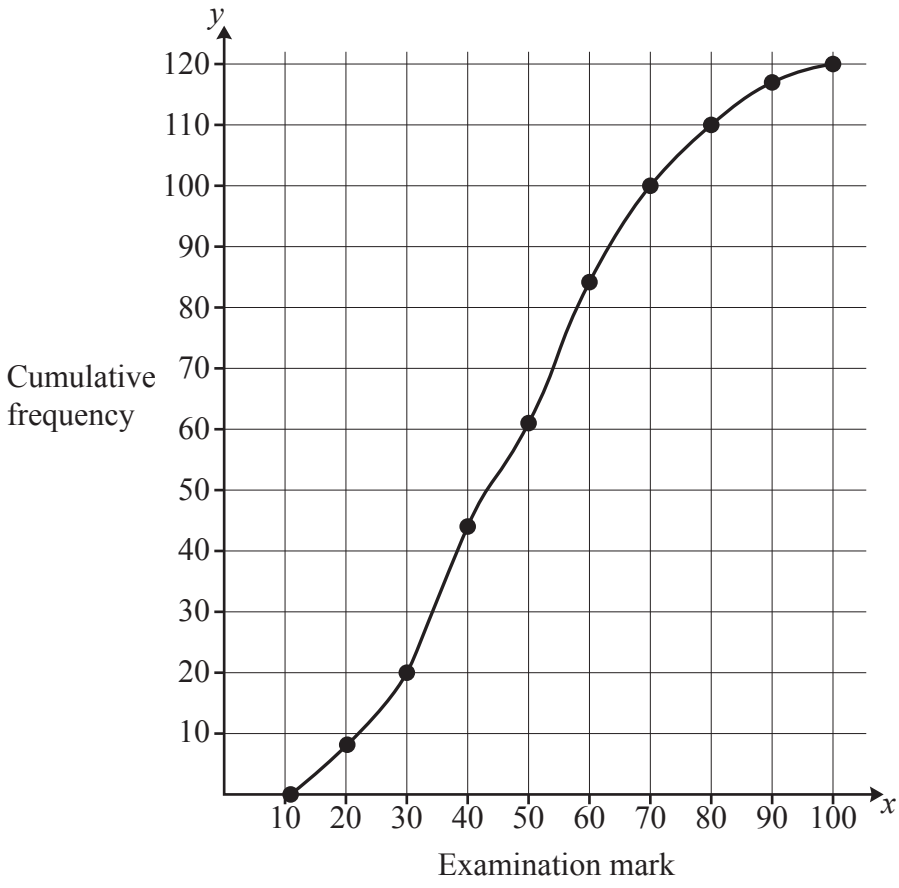
(a) It is proposed that the 30 pupils with the best marks in the examination are placed in the top set in Year 9.

Use the copy of the cumulative frequency curve in the **Printed Answer Booklet** to determine an estimate of the lowest mark needed for a pupil to be placed in the top set. **[2]**

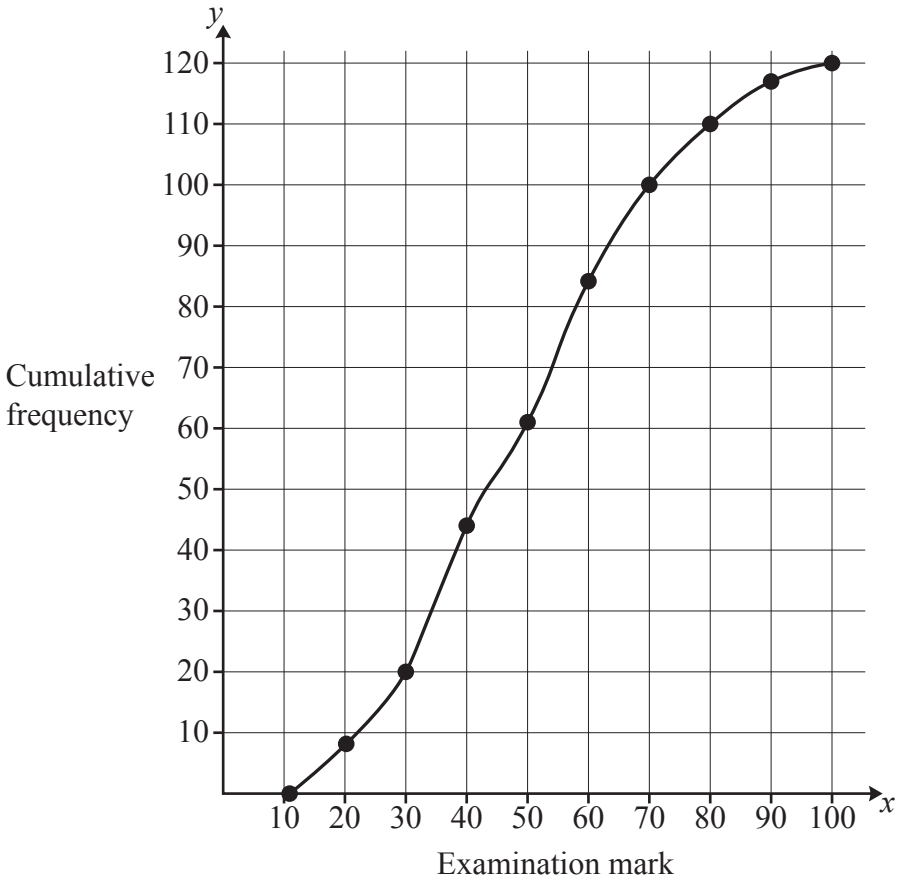
(b) The person who will teach the top set states that pupils will **not** manage in this class unless they obtain at least 75 marks in the examination.

Use the copy of the cumulative frequency curve in the **Printed Answer Booklet** to determine an estimate of how many pupils will be in the top set if the lowest mark is set at 75. **[2]**

8(a)


y
120
110
100
90
80
70
60
50
40
30
20
10
Cumulative frequency
10 20 30 40 50 60 70 80 90 100
x
Examination mark

8(b)


y
120
110
100
90
80
70
60
50
40
30
20
10
Cumulative frequency
10
20
30
40
50
60
70
80
90
100
x
Examination mark